

```
import pandas as pd
```

```
from google.colab import files
uploaded = files.upload()
df = pd.read_csv("airbnb.csv")
print(df.head())
print(df.columns)
```

Choose Files | airbnb.csv

airbnb.csv(text/csv) - 16181640 bytes, last modified: 28/5/2024 - 100% done

Saving airbnb.csv to airbnb (1).csv

Unnamed: 0	id	name
0	49849504	Perla bungalow
1	50891766	Authentic Beach Architect Sheltered Villa with...
2	50699164	cottages sataplia
3	49871422	Sapanca Breathable Bungalow
4	51245886	Bungalow Ev 2

	rating	reviews	host_name	host_id	address
0	4.71	64	Mehmetcan	357334205.0	Kartepe, Kocaeli, Turkey
1	New	0	Fatih	386223873.0	Kaş, Antalya, Turkey
2	4.85	68	Giorgi	409690853.0	Imereti, Georgia
3	5.0	13	Melih	401873242.0	Sapanca, Sakarya, Turkey
4	New	0	Arp Sapanca	414884116.0	Sapanca, Sakarya, Turkey

	features
0	2 guests,2 bedrooms,1 bed,1 bathroom
1	4 guests,2 bedrooms,2 beds,2 bathrooms
2	4 guests,1 bedroom,3 beds,1 bathroom
3	4 guests,1 bedroom,2 beds,1 bathroom
4	2 guests,1 bedroom,1 bed,1 bathroom

	amenities	price	country
0	Mountain view,Valley view,Lake access,Kitchen,...	8078	Turkey
1	Kitchen,Wifi,Dedicated workspace,Free parking ...	4665	Turkey
2	Mountain view,Kitchen,Wifi,Dedicated workspace...	5991	Georgia
3	Mountain view,Valley view,Kitchen,Wifi,Free pa...	11339	Turkey
4	Kitchen,Wifi,Free parking on premises,TV,Air c...	6673	Turkey

	bathrooms	beds	guests	toiles	bedrooms	studios	checkin
0	1	1	2	0	2	0	Flexible
1	2	2	4	0	2	0	4 00 pm - 11 00 pm
2	1	3	4	0	1	0	After 1 00 pm
3	1	2	4	0	1	0	After 2 00 pm
4	1	1	2	0	1	0	After 2 00 pm

	checkout
0	12 00 pm
1	10 00 am
2	12 00 pm
3	12 00 pm
4	12 00 pm

[5 rows x 23 columns]

```
Index(['Unnamed: 0', 'id', 'name', 'rating', 'reviews', 'host_name', 'host_id',
      'address', 'features', 'amenities', 'safety_rules', 'house_rules',
      'img_links', 'price', 'country', 'bathrooms', 'beds', 'guests',
      'toiles', 'bedrooms', 'studios', 'checkin', 'checkout'],
      dtype='object')
```

```
df["price"] = pd.to_numeric(df["price"], errors="coerce")
df = df.dropna(subset=["price"])
```

```
city_price = df.groupby("country")["price"].mean().reset_index()
property_price = df.groupby("features")["price"].mean().reset_index()
review_price = df.groupby("reviews")["price"].mean().reset_index()
```

```
print("Average Price by City:\n", city_price.head())
print("Average Price by Property Type:\n", property_price.head())
print("Average Price by Reviews:\n", review_price.head())
```

Average Price by City:

	country	price
0	Albania	4749.750000
1	Argentina	10647.000000
2	Armenia	6591.166667
3	Australia	26735.000000
4	Austria	20849.555556

Average Price by Property Type:

	features	price
0	1 guest,1 bed,0 bathrooms	19866.0
1	1 guest,1 bed,1 bathroom	9153.0
2	1 guest,1 bed,1 shared bathroom	8119.0

```

3 1 guest,1 bed,Shared toilet with sink 4343.0
4 1 guest,1 bedroom,0 bathrooms 1418.0
Average Price by Reviews:
reviews price
0 0 32462.055161
1 1 21335.424403
2 1,003 4013.000000
3 10 15565.576720
4 100 10231.529412

```

```

import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

from google.colab import files
uploaded = files.upload()

df = pd.read_csv("airbnb.csv")
df["price"] = pd.to_numeric(df["price"], errors="coerce")
df = df.dropna(subset=["price"])
X = df[["country", "features", "reviews"]]
y = df["price"]
if not pd.api.types.is_numeric_dtype(X["reviews"]):
    cat_features = ["country", "features", "reviews"]
else:
    cat_features = ["country", "features"]

preprocessor = ColumnTransformer(
    transformers=[("cat", OneHotEncoder(handle_unknown="ignore"), cat_features)],
    remainder="passthrough"
)
model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", LinearRegression())
])
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model.fit(X_train, y_train)

print("Training R^2:", model.score(X_train, y_train))
print("Test R^2:", model.score(X_test, y_test))

```

[Choose Files](#) `airbnb.csv`

airbnb.csv(text/csv) - 16181640 bytes, last modified: 28/5/2024 - 100% done
 Saving airbnb.csv to airbnb (2).csv
 Training R^2: 0.668608975901996
 Test R^2: 0.15227275366007154

```

from sklearn.linear_model import Ridge
model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", Ridge(alpha=1.0))
])
from sklearn.model_selection import GridSearchCV

params = {"regressor__alpha": [0.1, 1.0, 10.0, 100.0]}
grid = GridSearchCV(model, params, cv=5)
grid.fit(X_train, y_train)

print("Best alpha:", grid.best_params_)
print("Best R^2:", grid.best_score_)

```

Best alpha: {'regressor__alpha': 10.0}
 Best R^2: 0.1405370023758516

```

import numpy as np

feature_names = grid.best_estimator_.named_steps["preprocessor"].get_feature_names_out()
coefs = grid.best_estimator_.named_steps["regressor"].coef_

importance = pd.DataFrame({"feature": feature_names, "coef": coefs})
print(importance.sort_values("coef", ascending=False).head(10))

```

	feature	coef
328	cat__features_10 guests,5 bedrooms,7 beds,5 ba...	153760.402082
895	cat__features_16 guests,19 bedrooms,18.5 bathr...	103788.654669
850	cat__features_16 guests,12 bedrooms,12 beds,12...	83998.291013
809	cat__features_16 guests,10 bedrooms,16 beds,8 ...	65099.804283

109	cat__country_ United Arab Emirates	65038.489397
90	cat__country_ Seychelles	64513.366496
320	cat__features_10 guests,5 bedrooms,6 beds,5 ba...	62549.208083
598	cat__features_13 guests,8 bedrooms,1 bed,9 bat...	57597.745059
1082	cat__features_16 guests,7 bedrooms,7 beds,7 ba...	56782.863336
835	cat__features_16 guests,11 bedrooms,14 beds,8 ...	53311.238641

```
import pandas as pd
from sklearn.linear_model import Ridge
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from google.colab import files

uploaded = files.upload()
df = pd.read_csv("airbnb.csv")

df["price"] = pd.to_numeric(df["price"], errors="coerce")
df["reviews"] = df["reviews"].astype(str).str.replace(", ", "").astype(float)
df = df.dropna(subset=["price", "reviews"])

X = df[["country", "features", "reviews"]]
y = df["price"]

preprocessor = ColumnTransformer(
    transformers=[("cat", OneHotEncoder(handle_unknown="ignore"), ["country", "features"])],
    remainder="passthrough",
    force_int_remainder_cols=False
)

model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", Ridge(alpha=1.0))
])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model.fit(X_train, y_train)

print("Training R^2:", model.score(X_train, y_train))
print("Test R^2:", model.score(X_test, y_test))
```

Choose Files | airbnb.csv

airbnb.csv(text/csv) - 16181640 bytes, last modified: 28/5/2024 - 100% done
 Saving airbnb.csv to airbnb (4).csv
 Training R^2: 0.5570743596545874
 Test R^2: 0.20782234874283811

```
from sklearn.model_selection import GridSearchCV

params = {"regressor__alpha": [0.1, 1.0, 10.0, 100.0]}
grid = GridSearchCV(model, params, cv=5)
grid.fit(X_train, y_train)

print("Best alpha:", grid.best_params_)
print("Best CV R^2:", grid.best_score_)
print("Test R^2:", grid.score(X_test, y_test))
```

Best alpha: {'regressor__alpha': 10.0}
 Best CV R^2: 0.12257259042619786
 Test R^2: 0.1732930587601751

```
import pandas as pd
from sklearn.linear_model import Ridge
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from google.colab import files
from ipywidgets import interact

uploaded = files.upload()
df = pd.read_csv("airbnb.csv")

df["price"] = pd.to_numeric(df["price"], errors="coerce")
df["reviews"] = df["reviews"].astype(str).str.replace(", ", "").astype(float)
df = df.dropna(subset=["price", "reviews"])

X = df[["country", "features", "reviews"]]
y = df["price"]

preprocessor = ColumnTransformer(
```

```

transformers=[("cat", OneHotEncoder(handle_unknown="ignore"), ["country", "features"])],
remainder="passthrough",
force_int_remainder_cols=False
)

model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", Ridge(alpha=1.0))
])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model.fit(X_train, y_train)

def predict_price(country, features, reviews):
    sample = pd.DataFrame({
        "country": [country],
        "features": [features],
        "reviews": [reviews]
    })
    pred_price = model.predict(sample)[0]
    print(f"Suggested Price: ${pred_price:.2f}")

interact(
    predict_price,
    country=df["country"].unique(),
    features=df["features"].unique(),
    reviews=(0, 500, 10)
)

```

Choose Files **airbnb.csv**

airbnb.csv(text/csv) - 16181640 bytes, last modified: 28/5/2024 - 100% done
 Saving airbnb.csv to airbnb (5).csv

country

features

reviews

Suggested Price: \$232.75

```

predict_price
def predict_price(country, features, reviews)

```

<no docstring>

pip install ipywidgets

```

Requirement already satisfied: ipywidgets in /usr/local/lib/python3.12/dist-packages (7.7.1)
Requirement already satisfied: ipykernel>=4.5.1 in /usr/local/lib/python3.12/dist-packages (from ipywidgets) (6.17.1)
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Requirement already satisfied: ipython>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from ipywidgets) (7.34.0)
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Requirement already satisfied: parso<0.9.0,>=0.8.4 in /usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython>=4.0.0) (0.8.4)
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Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-packages (from prompt-toolkit!=3.0.0,!<3.0.1,<3.1.0->ipython>=4.0.0) (0.2.13)
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Requirement already satisfied: tinycss2<1.5,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->n
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat->n
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsons

```
import ipywidgets as widgets
from IPython.display import display
```

```
slider = widgets.IntSlider(value=50, min=0, max=100, step=5, description="Test Slider:")
display(slider)
```

Test Slider:  40

```
import pandas as pd
import numpy as np
from sklearn.linear_model import Ridge
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

data = {
    "country": ["US", "US", "IN", "IN", "UK", "UK"],
    "features": ["Apartment", "Villa", "Apartment", "Villa", "Apartment", "Villa"],
    "reviews": [120, 45, 300, 150, 80, 60],
    "price": [200, 350, 80, 150, 180, 300]
}
df = pd.DataFrame(data)

X = df[["country", "features", "reviews"]]
y = df["price"]

preprocessor = ColumnTransformer(
    transformers=[("cat", OneHotEncoder(handle_unknown="ignore"), ["country", "features"])],
    remainder="passthrough"
)

model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", Ridge(alpha=1.0))
])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model.fit(X_train, y_train)

def rule_based_adjustment(base_price, reviews):
    if reviews > 200:
        return base_price * 1.1
    elif reviews < 50:
        return base_price * 0.9
    return base_price

def suggest_price(country, features, reviews):
    sample = pd.DataFrame({
        "country": [country],
        "features": [features],
        "reviews": [reviews]
    })
    base_price = model.predict(sample)[0]
    final_price = rule_based_adjustment(base_price, reviews)
    return round(final_price, 2)

print("US Apartment, 120 reviews → $", suggest_price("US", "Apartment", 120))
print("IN Villa, 300 reviews → $", suggest_price("IN", "Villa", 300))
print("UK Apartment, 20 reviews → $", suggest_price("UK", "Apartment", 20))
```

US Apartment, 120 reviews → \$ 174.53
IN Villa, 300 reviews → \$ 111.23

